

Ganeshman Singh Multiple Campus
Pre-Board Examination
Subject: Data Structure and Algorithm

Full Marks: 60

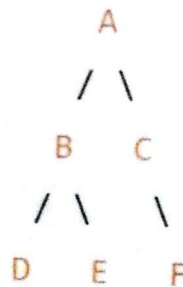
Time: 3 hours

Pass Marks: 24

[6x5=30]

Attempt any six questions.

1. Define data structure. Explain the difference between primitive and non-primitive data structures with suitable examples.
2. Define stack. Explain the concept of LIFO with a real-life example.
3. Define a queue. How is it different from a stack? Explain the **FIFO** principle with a real-life example.
4. What is a graph? Explain the difference between BFS and DFS algorithms with an example.
5. What is a singly linked list? Write the algorithm to delete the first node of a singly linked list and explain it with a diagram.
6. Write the algorithm for the Bubble Sort technique. Sort the following list using Bubble Sort and show the steps:
[45, 23, 11, 89, 77]
7. Write the in-order, pre-order, and post-order traversals for the following binary tree:



Attempt any two questions.

[2x10=20]

8. What is a Binary Search Tree (BST)? Write algorithms for insertion and deletion of a node in a BST. Draw the BST after inserting the following keys in order:
50, 30, 70, 20, 40, 60, 80
Then delete the node with key 30 and show the resulting tree.
9. Explain the Divide and Conquer strategy used in Merge Sort. Write the detailed algorithm for Merge Sort and analyze its time complexity. Also, perform Merge Sort on the array:
[38, 27, 43, 3, 9, 82, 10]
showing each step clearly.
10. Define graph and tree data structure. Explain breadth first traversal and depth first traversal with example.

Attempt all the MCQ questions. [10x1= 10]

Circle (O) the correct answer.

1. What is the main advantage of a linked list over an array?
 - a) Fixed size
 - b) Faster access to elements
 - ☒ c) Dynamic size
 - d) Random access
2. Which of the following data structures allows insertion and deletion only at one end?
 - a) Queue
 - ☒ b) Stack
 - c) Deque
 - d) Circular Queue

G. M. S. MULTIPLE CAMPUS

Kalanki, Kathmandu, Nepal

Pre-board Examinations – Jestha, 2082

Subject: Probability and Statistics

Level: Bachelor in Computer Application / Third Semester (BCA)

Time: 3 hrs

Full Marks: 60

Code No: CAST 202

Name: _____ Symbol No: _____

Candidates are required to give their answers in their own words as far as practicable. The figures in the margin indicate full marks.

Group B

Attempt any SIX questions.

[6×5 = 30]

- Describe scope and limitation of Statistics.
- Determine average wages from following data:

Wages	25-30	30-35	35-40	40-45	45-50	50-55	55-60	60-65	65-70
No. of Worker	10	13	18	21	24	28	20	11	8

- Calculate Karl Pearson's correlation coefficient from the following data:

Sales	43	41	36	34	50
Expenses	10	22	13	19	17

- Estimate the marks in JAVA when the marks in Statistics is 65 by using following data:

Marks in Statistics	57	58	59	59	60	61	62	64
Marks in JAVA	77	78	75	78	82	82	79	81

- Fit Binomial Distribution from the following data where $p = 0.5$

No. of heads	0	1	2	3	4
Frequency	28	62	46	20	4

- How do you determine sample size in sampling? Explain briefly.
- Write short notes on simple random sampling.

Group C

Attempt any TWO questions.

[2×10 =

20]

9. Student's age in the regular daytime BCA program and the morning time BCA program of a campus are described by two samples. If the homogeneity in age of the class is positive factor in learning make suggestion, with reason, which of two groups will be easier to teach?

Regular BCA program		Morning BCA program	
Age	No. of Students	Age	No. of Students
23	9	27	10
29	2	31	8
28	5	30	5
22	10	29	4
30	1	28	6
21	4	33	5
25	11	34	5
26	6	35	11
27	3	36	2
24	9	32	4
Total	60	Total	60

10. Given a normal distribution with mean 200 and s.d. 20, find the probability that

i) $P(X > 180)$

ii) $P(X < 220)$

iii) $P(160 < X < 240)$

iv) $P(X > 220)$

v) $P(X < 180 \text{ or } X > 220)$

vi) 10% of the values are less than what values of X?

11. The labor productivity indexes of Nepal are recorded as below:

Sector	Year		
	2015	2016	2017
Agriculture	100	125	138
Manufacturing	100	60	53
Community and Social service	100	89	80

Does the labor productivity index vary due to the;

- i) difference in the sector ii) difference in the time period?

G. M. S. MULTIPLE CAMPUS

Kalanki, Kathmandu, Nepal

Terminal Examinations – Jestha, 2082

Subject: System Analysis and Design

Level: Bachelor Level / Third Semester (BCA) Time: 3 hrs

Candidates are required to give their answers in their own words as far as practicable. The figures in the margin indicate full marks.

Attempt any two

Group A 2x10=20

1. Define information System? Why do we need it? Discusses different types of information system with suitable example.
2. What is DFD? Discuss different level of DFD with suitable example.
3. What is system implementation? Discuss different activities of system implementation in detail.

Group B 6x5=30

Attempt any six

1. Describe and compare the various source of software.
2. What different skill do you need to be a good system analysis?
3. Describe the steps involved in corporate strategic planning?
4. Describe the steps in creating decision table? How do we reduce the size and complexity of a decision table?
5. What is information gathering? Discuss questionnaire along with its merits and demerits.
6. Explain Different types of software testing?
7. How can you transform ER Diagram into relation? Explain with your own suitable example.

Ganeshman Singh Multiple Campus
Term Exam- 2082

Subject: OOP in Java

FM: 60

Group B – Short Answer Questions

Attempt any six questions. (6 × 5 = 30 Marks)

- 1) Explain the difference between method overloading and method overriding with examples.
- 2) What is the role of the final keyword in Java? Explain with an example.
- 3) Write a Java program to find the factorial of a number using recursion.
- 4) What is the difference between abstract class and interface?
- 5) Describe the use of this and super keywords with examples.
- 6) Explain exception handling in Java using try, catch, and finally.
- 7) Differentiate between object and class with real-world examples.
- 8) Write a short note on:
 - a) Access Modifier in Java.
 - b) JVM and JRE

Group C – Long Answer Questions

Attempt any two questions. (2 × 10 = 20 Marks)

- 9)
a) Design a Java class BankAccount with the following specifications:

- Data members: accountNumber, accountHolder, balance
- Member methods:
 - deposit(amount) – adds amount to balance
 - withdraw(amount) – deducts amount if sufficient balance exists
 - displayDetails() – prints account details

Write a program to demonstrate all functionalities.

b) Explain the concept of constructor overloading in Java. Write a program that defines a Student class with multiple constructors for different initialization needs (e.g., roll number only, roll number and name, all details).

10)

a) What is polymorphism? Describe the two types of polymorphism in Java:

- Compile-time polymorphism with method overloading
- Run-time polymorphism with method overriding

Support your explanation with Java code examples.

b) Discuss dynamic method dispatch in Java. Write a program where a superclass reference is used to call overridden methods in subclass objects. Explain the output to show how polymorphism works at runtime.

11)

a) Explain inheritance in Java. Create a superclass Employee and a subclass Manager that inherits from Employee. Demonstrate:

- Method overriding
- Use of super keyword

b) Write a Java program to demonstrate multilevel inheritance. Use classes Person, Employee, and Manager. Show how data flows across multiple levels of the hierarchy and how methods are reused or overridden.